Electric Vehicle Market Performance & Insights

Ideation and Problem Statement

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

Problem: EV data can contain many manufacturers, model years, vehicle categories and technical/commercial attributes. Without a repeatable cleaning process and clear visual summaries, comparing records is time-consuming and can lead to misleading conclusions. The project creates a reproducible analytics workflow and an interactive way to explore the available records.

Detailed Project-Specific Content

- Aim: clean and structure EV records, explore attributes, and communicate descriptive patterns through Tableau and Flask.
- Objectives: document the dataset; clean it with Python/Pandas; retain a cleaning report; use SQL/SQLite for structured queries; compare manufacturers, years and categories; explore battery type/capacity, range, price and rating where fields support it; publish a Tableau dashboard/story; and provide Overview, Dashboard, Story and Methodology web pages.
- Target users: students, instructors and people exploring EV product attributes. The project is not a certified vehicle-rating system, sales forecast, battery-safety evaluation or life-cycle emissions model.
- Scope: descriptive analysis of the supplied dataset. Market-wide claims require proof of representativeness, field meaning and source coverage.
- Expected deliverables: source code, cleaned CSV, cleaning report, SQL/database assets, Tableau visualizations, Flask interface, test evidence, user guide and final report.
- Success criteria: data preparation is repeatable; outputs reconcile; metric definitions and units are visible; dashboard/story links work; web routes render; and limitations are stated clearly.
- Risks: uncertain data provenance, ambiguous units, biased/missing records, broken embeds and unverified deployment. Controls include data dictionary, independent metric checks, fallback links and explicit caveats.

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.